Reforming Power Electronics Education in the Era of AI: A Pilot Course by the University of Arkansas Power Group

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Abstract:

As artificial intelligence (AI) transforms modern society, power electronics plays a dual role: it powers the AI era, and it is increasingly being transformed by AI itself. Yet current PE education does not typically introduce AI as a systematic engineering toolbox, and generic AI education cannot simply be transplanted into the PE context. To help bridge this educational gap, the University of Arkansas launched a first-of-its-kind graduate course, *Fundamentals of AI for Power Electronics*, which combines AI fundamentals with PE-specific applications through vibe programming, hands-on implementation, and project-based learning. To extend its impact beyond the classroom, the course is further supported by open-source code materials, an algorithm selector web app, and a customized AI agent. The message is simple but urgent: preparing AI-ready power electronics engineers should become a shared priority for the entire power electronics community.

Section I. Introduction

Power electronics (PE) is both a critical enabler of the artificial intelligence (AI) era and a major beneficiary of AI itself. On one hand, before AI systems are limited by computation, they are often limited by power: power delivery, conversion efficiency, thermal management, power density, and energy utilization all directly shape the practical scalability of modern computing infrastructure. From data centers and AI accelerators to edge devices, electric transportation, and renewable-energy-powered grids, advances in AI ultimately depend on advances in PE [1]. On the other hand, AI is increasingly becoming a foundational intelligence layer for PE, offering a diverse toolbox for modeling, design, control, and maintenance of power converters [2]. In this sense, the relationship between AI and PE is deeply reciprocal: PE sustains AI physically, while AI accelerates the evolution of PE technologically [3].

This reciprocal relationship creates a timely educational challenge. Compared with software-native disciplines, PE is fundamentally a hardware-native field rooted in physical laws and constraints. Traditional PE training emphasizes physical intuition, first-principles analysis, and the ability to reconcile theory with hardware behavior. These characteristics make PE education inherently different from the largely data- and software-centric environment in which AI is typically taught.

While AI is increasingly discussed in research and industry, it is still rarely introduced in PE curricula as a systematic set of generic, high-performance tools that can augment engineering reasoning and workflow efficiency [4]. Conversely, fundamental AI education that is designed around canonical data types and benchmark problems cannot simply be transplanted into PE classrooms without adaptation [5]. PE-unique data modalities include semi-structured waveform and spectral data, multi-physics field distributions, graph-like interconnections such as PCB layouts and power module structures, etc. Teaching AI for PE therefore requires more than explaining algorithms; it requires showing how those algorithms interact with engineering priors and physical compliance.

Several examples illustrate why this integration is nontrivial. In PE design and modeling, students must learn how to incorporate domain knowledge to choreograph AI-based data-driven workflows rather than treating the model as a purely black-box predictor. In many cases, AI tools must also help bridge the persistent deviation between analytical equations or circuit simulation and experimental measurements [6]. In PE control, beyond accuracy, AI-based controllers must also satisfy stringent constraints on latency and memory footprint for online, real-time implementation [7]. In maintenance-related applications, such as fault diagnosis and failure mode and effects analysis, labeled data are often scarce because data acquisition is destructive, time-consuming, and resource-intensive [8]. Therefore, students must understand how to build AI solutions that generalize reliably despite falling under such low-data conditions. These issues are central to practical AI deployment in PE, yet they are seldom addressed in conventional AI education.

It is urgent for cross-disciplinary education that prepares the next generation of AI-ready PE engineers, while supporting upskilling of the existing workforce. Responding to this need, the University of Arkansas (UA) developed what is, to the best of our knowledge, the first graduate-level course dedicated to the fundamentals of AI for PE. The course was designed to introduce prevalent AI algorithms through the lens of PE applications, combining conceptual instruction with practical case studies spanning multiple phases of the PE lifecycle. Beyond traditional lectures, the course includes interactive coding sessions, hands-on implementation, project-based learning, and evidence-based reasoning. It also introduces vibe programming concepts for the first time to PE education, enabling students to explore “zero-line-of-code” AI development workflows using generative AI tools, which is particularly important because many students entering the course have little or no prior background in AI.

To further lower the barrier to entry and broaden the impact of this educational effort, we also developed a supporting ecosystem around the course. The hands-on code materials have been open-sourced on GitHub[1] [9], and we additionally created an interactive web application[2] that guides users through step-by-step algorithm selection for PE scenarios, addressing a common challenge faced by beginners: choosing an appropriate AI method for a given PE problem. We further built a bespoke AI agent[3] to align AI recommendations with available open-source materials that explains key concepts in a more accessible manner, which serves not merely as a chatbot, but as a personalized educational assistant that shortens the path from curiosity to implementation and supports self-paced learning.

In this article, we present the philosophy, early experience, and supporting educational tools associated with this course initiative. Looking ahead, we also offer recommendations for the broader PE community, including educators, students, industry practitioners, and government agencies, on how to collaboratively build the next generation of AI-integrated PE education and workforce development.

Section II. Education Efforts at the UA: Fundamentals of AI for Power Electronics

To address the growing need for AI-ready PE engineers, UA power group launched a graduate-level course, *Fundamentals of AI for Power Electronics*, in the Fall 2025 semester. Led by Dr.

- Supplementary GitHub code repository:
[1]https://github.com/XinzeLee/Fundamentals_of_AI_for_PE

- Interactive web application to select AI algorithms for PE scenarios
[2]https://github.com/XinzeLee/AI_for_PE_Algorithm_Selector

- Bespoke AI agent to guide AI algorithm selection for PE scenarios
[3]https://chatgpt.com/g/g-698618895c2481919e113c49bafe23ee-fundamentals-of-ai-for-pe

Xinze Li as the instructor, with Dr. Alan Mantooth providing leadership and support, this course represents a pioneering educational effort to systematically introduce AI into PE graduate training.

The inaugural offering enrolled 14 students, including 3 M.S. students and 11 Ph.D. students, with 2 additional sit-in students. The cohort was interdisciplinary: 11 students had PE backgrounds, while others came from cybersecurity, power semiconductor devices and modeling, and electronic design automation. Most students entered the course with little (3 students) or no prior exposure (13 students) to AI. The course is intended to prepare the next generation of PE workforce rather than AI engineers.

Fig. 1. Group photo of the graduate course "*Fundamentals of AI for Power Electronics*" at the UA; the instructor is the sixth person from the left.

- **Main Philosophies**

Several philosophies guided the new course. First, AI is treated as a set of engineering tools rather than as a purely algorithmic or software programming discipline. For PE students, the key question is often not how to build every AI model from scratch, but how to select, tune, and evaluate AI tools effectively for PE tasks. To this end, the course introduced "vibe programming" into PE education for the first time, encouraging students to use generative or agentic AI tools, such as GPT models, Cursor, and Claude Code, to enable "zero-line-of-code" or low-code AI implementation. This approach lowered the entry barrier for students with limited coding or AI backgrounds, while reinforcing an important lesson: generative AI is only effective when guided by strong conceptual understanding, good prompts, and sound PE domain judgment.

Second, the course emphasizes concept-driven and evidence-based learning. Core AI concepts were prioritized over programming details so that students could understand why an AI algorithm succeeds or fails in a given PE scenario. Hands-on experiments were used extensively to help students compare algorithms, understand trade-offs, and interpret outcomes.

Third, project-based learning is applied. Homework assignments require students to implement AI algorithms on benchmark datasets and downstream PE problems. The midterm exam was designed as a hackathon challenge on practical PE tasks, in which students trained data-driven machine learning (ML) models for magnetic core loss modeling that are agnostic to magnetic flux density waveforms leveraging a down-sampled MagNet dataset [10]. The final project included both an investigation of how AI could benefit their own research and a group effort to develop customized PE-GPT agents [11]. In this capstone experience, students put all their acquired knowledge to the test: they combine large language models with tools, knowledge, memory, and workflow logic to build task-oriented AI agents for their own PE applications.

A new teaching format was also explored: interactive coding sessions. In certain lectures, students are given partially completed Jupyter Notebooks and guided through experiments step by step in class, with the instructor filling in missing blocks, running code, and interpreting results on the run. Completed master-version Jupyter Notebooks are then shared after the class. This format allows students to learn AI concepts and coding practices whereas reducing the intimidation often associated with first exposure to AI implementation.

■ A Glimpse of Course Syllabus

The 16-week course syllabus was organized to span both AI fundamentals and representative PE applications, following the sequence shown in Table 1. In this way, the course moved from foundational algorithms to increasingly PE-specific and application-oriented topics.

Table 1. Syllabus of the Graduate-level Course: Fundamentals of AI for PE

Module 1	Introduction to AI in PE
Module 2	Single-objective metaheuristic optimization for converters
Module 3	Multi-objective optimization
Module 4	Fundamentals of ML for PE
Module 5	Ensemble learning
Module 6	Fundamentals and good practices of neural networks (NNs)
Module 7	Advanced NNs for time-series data and probabilistic regression
Mid-term exam	**AI hackathon: NN-based magnetic core loss modeling using a down-sampled MagNet dataset**
Module 8	AI-augmented parameter design for power converters
Module 9	One-stop AI for modulation design of dual-active-bridge converters
Module 10	Generative AI for PE process automation
Final group project begins	**Build your own agentic AI (customized PE-GPT) for PE design tasks of your choice**
Module 11	Physics-informed ML for PE modeling and design
Module 12	AI in PE control

■ Course Outcome and Student Feedback

The early outcomes of the course are encouraging. In the mid-term exam for magnetic core loss modeling, as shown in Fig. 2, a student significantly improved the accuracy of the Steinmetz equation from $R^2 = 0.926$ with over 15% mean average percentage error (MAPE) to $R^2 = 0.99$ with less than 4.5% MAPE. More broadly, students progressed from having almost no AI background to building diverse PE-GPT agents [11] for practical PE design tasks, including PCB transformer design with automated Ansys multi-physics verification, inductor design for dc-dc converters, integrated device selection and transformer synthesis for dual-active-bridge converters, magnetic component modeling, and automation of double-pulse-test LTspice simulation and gate-driver design[4].

Selected student feedback listed in Table 2 also reflects the value of the course. Survey results indicated that students felt more confident in applying AI to PE problems and believed that the

■ Students' customized PE-GPT project outcome video:
[4]https://www.linkedin.com/feed/update/urn:li:activity:7405449237741756416/

course filled an important gap in current PE education, as shown in Fig. 3. Many reported that the course fundamentally changed how they viewed generative AI tools such as ChatGPT, shifting from conservative resistance to purposeful AI development. Others highlighted the intuitive, PE-focused explanation of AI concepts, the practical relevance of optimization and data-driven modeling, and the final generative AI group project as valuable parts of the learning experience.

Table 2. Selected Student Feedback

Enrolled Students and their Backgrounds	Please share your comments on 1) your experience with this new course, and 2) how the course may influence your future research
Daixin Chen • 2nd-year Ph.D. student • Supervisor: Dr. Xiaoqing Song • Research focus: DC circuit breakers	I found this course both challenging and inspiring. As someone with no prior background in AI, I initially felt that the material was difficult because it introduced a cutting-edge field with detailed, rigorous, and highly enriching content. However, as the course progressed, I increasingly realized how valuable it is for power electronics research. It not only helped me understand AI methods in the context of PE, but also reshaped how I think about my research workflow and problem-solving approach. By integrating the knowledge, skills, and methodologies from this course into my current work, I gained a new perspective on how to address PE problems more efficiently and systematically. In particular, the final project of building a PE-GPT was the most exciting part for me, because it showed me how deeply AI can be incorporated into my work and how much it can enhance research productivity. I truly believe this is an excellent course, and I would be very excited to recommend it to anyone in the power electronics community.
Eric Allee • 5th-year Ph.D. student • Supervisor: Dr. Yue Zhao • Research focus: medium-voltage SiC-based inverter design	I went into this course with no real expectations and no background in the foundational topics that underpin modern AI. There were moments where I felt genuinely overwhelmed, but the material was presented in a way that made it possible to build a basic understanding quickly and still complete the work meaningfully. What surprised me most was how much the course demystified AI; by breaking it down into its underlying methods and showing how the pieces fit together, it changed how I think about it entirely. I came in viewing AI as magic and left with the confidence to build my own tools. I have since developed custom programs with embedded AI agents to streamline the processing of large experimental datasets, work that would have taken far longer manually. I would not have attempted any of it without this course.
Md Majharul Islam • 3rd-year Ph.D. student • Supervisor: Dr. Alan Mantooth • Research focus: compact model development for semiconductor devices	This course provided a strong foundation in building LLM-based agents from scratch, including their core components, ML algorithms and techniques for customization and domain-specific tuning. I gained practical exposure to multi-agent systems, Retrieval-Augmented Generation (RAG), and the integration of AI into electronic design automation (EDA) workflows, including invoking local simulation and CAD tools to solve specific tasks and making next decision based on the results. By applying these concepts, I have prototyped and continue to develop a co-pilot–style EDA agent to support my research in semiconductor device study, simulation, netlist and layout generation, and design rule checking. This work has significantly improved both my overall understanding and the integration of AI into my research for increasing productivity. Overall, the course deepened my understanding of AI systems, sharpened my skills, and strongly influenced blending my research in semiconductor device modeling and circuit design leveraging an Agentic Approach.
Wesley Schwartz • 3rd-year Ph.D. student • Supervisor: Dr. Chris Farnell • Research focus: cybersecurity for critical infrastructure	With generative AI and LLMs becoming a mainstream tool in the workforce today, it is increasingly important that power electronics students and engineers understand how AI can be used effectively to assist with everyday engineering tasks. The Fundamentals of AI for PE course at the UofA was a great foundational course covering not only the fundamentals of AI but also its practical applications. Developing a PE-GPT agent in the final project was both challenging and rewarding, and it offered a new perspective on how AI can meaningfully enhance our research in power electronics. Nearly every topic in this course was a skill that I can apply to either my research on edge-AI for critical infrastructure, or my work as a Test Engineer at the National Center for Reliable Electric Power Transmission.

**The student quotations included in this article are reprinted with permission from the students.*

Overall, this UA educational experiment suggests that AI education for PE should not simply import generic AI content into a hardware-native discipline. Instead, it should be redesigned around PE-specific data, engineering workflows, and applications. When taught in this way, AI can become accessible even to students with little prior background, while remaining rigorous enough to support meaningful research and industry innovation.

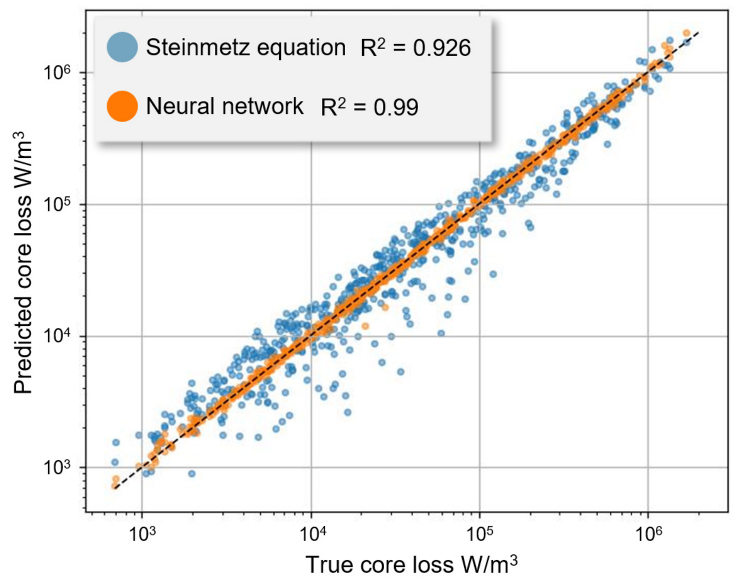

Fig. 2. Midterm exam results from an enrolled Ph.D. student: core-loss modeling using a downscaled MagNet Challenge dataset.

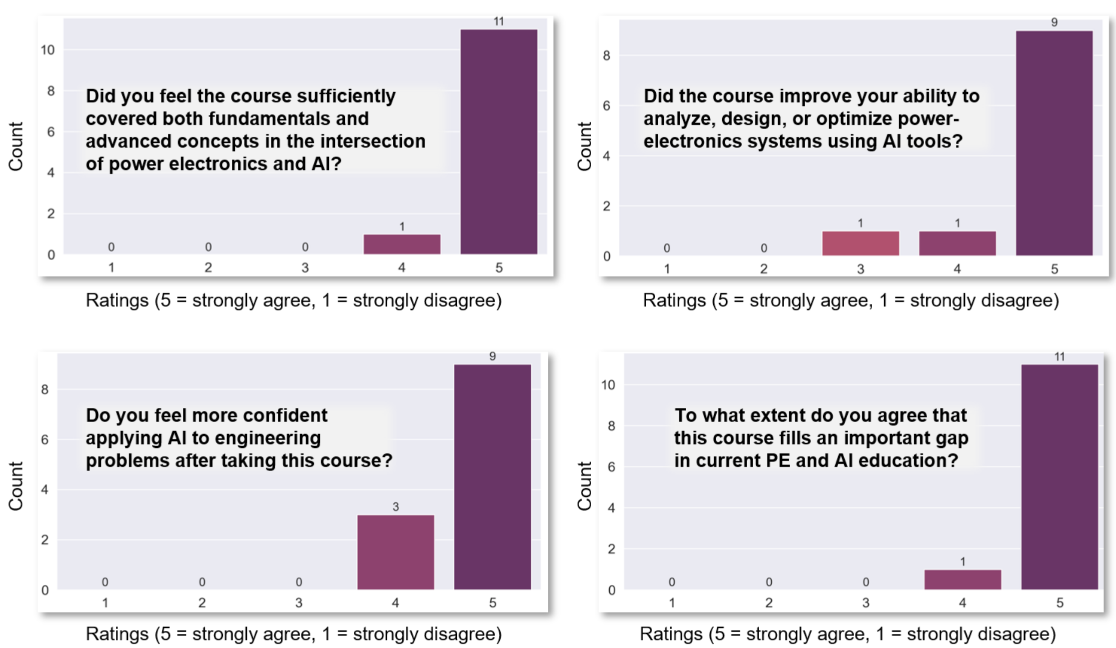

Fig. 3. Summary of the student survey and selected student feedback.

Section III. Available Supplementary Materials and Suggested Usage

To make AI for PE more accessible, actionable, and scalable, a set of supplementary materials was developed to support self-paced learning, rapid hands-on practice, and classroom adoption.

These materials were designed for researchers, educators, and industry practitioners who wish to explore AI for PE in a more practical and structured manner.

■ Available Resources

Three complementary types of resources are currently available, as shown in Fig. 4. These materials are actively maintained to include more AI algorithms and PE case studies.

First, selected hands-on code materials have been open-sourced [9] to demonstrate good practices for implementing AI in PE and to facilitate conceptual understanding, spanning both AI fundamentals and downstream PE case studies. In total, the GitHub repository currently includes more than 10,000 lines of code, 26 Jupyter Notebooks, 6 PE datasets, several representative PE case studies, and implementations of 24 AI algorithms, covering 9 sections in total.

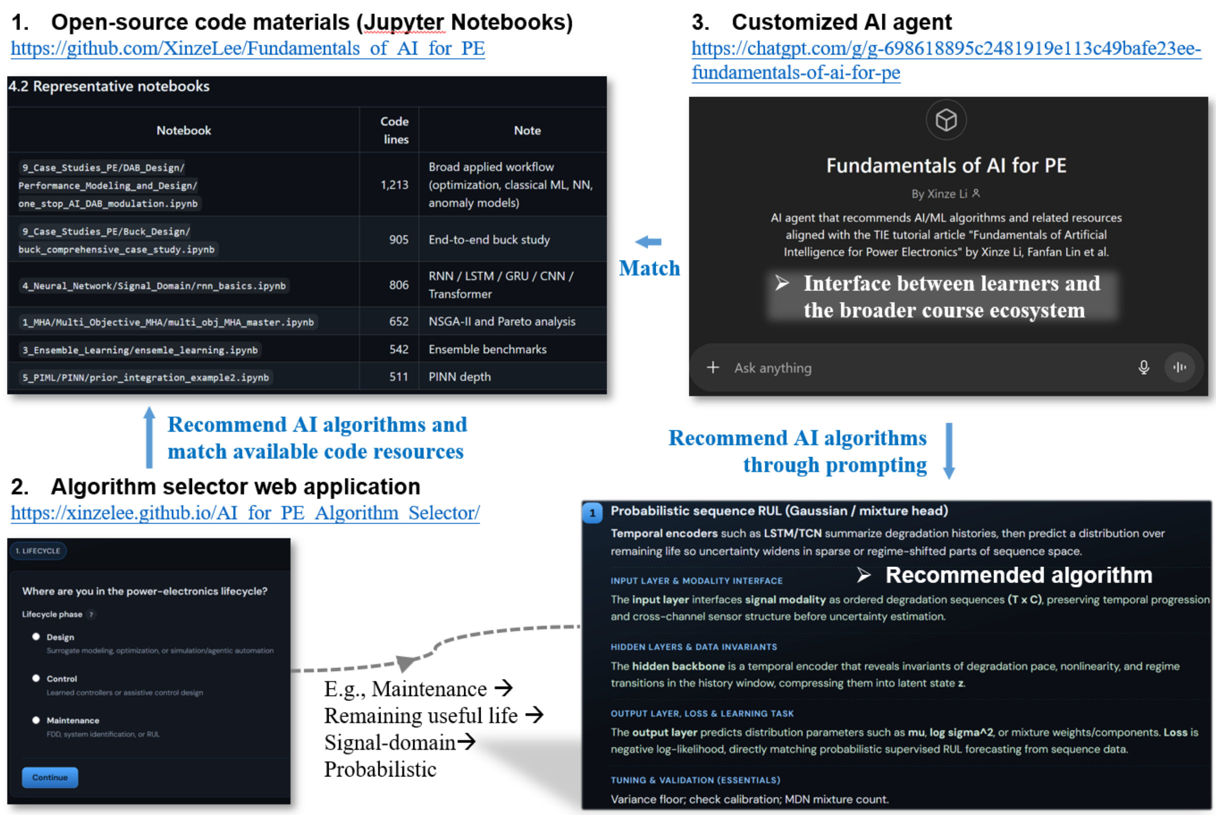

Fig. 4. Supplementary materials for the educational initiative, *Fundamentals of AI for Power Electronics*.

Second, an algorithm selector web application has been developed to facilitate interactive, step-by-step algorithm selection and to connect users with relevant code materials. The app adopts a tree-style process to narrow AI choices based on specific feature descriptions of PE tasks. It first asks the user to identify the PE lifecycle phase, design, control, or maintenance, and then branches accordingly. For PE design, it selects among modeling, optimization, and process automation; for PE control, it separates AI as the controller from AI-assisted control design; and for maintenance, it includes fault detection and diagnosis, system identification, and remaining useful life prediction. The app then refines the recommendation using problem-specific descriptors such as input/output data modality (tabular, signal, graph, field, or hybrid), labelled data availability (abundant or scarce), optimization dimensionality, single- versus multi-objective formulation, continuous versus discrete action/search space, supervision level, and deployment constraints such as online execution. Based on this sequence, the app maps the PE problem to a corresponding recommendation path and returns candidate algorithms, practical notes, and matched Jupyter Notebooks.

Third, a customized AI agent was developed as an interface layer between learners and the broader course ecosystem. The agent reduces the friction associated with navigating fragmented resources, terminology, and implementation pathways in AI for PE. For beginners, it helps translate high-level engineering goals into actionable next steps; for more advanced users, it can accelerate exploration by connecting relevant AI concepts and algorithms, examples, and code materials in a unified and coherent workflow.

■ A Quick Guide for Using these Materials

These resources can be used in different ways depending on the user's goal.

If the immediate need is to obtain a quick algorithm recommendation, the algorithm selector web app or the customized AI agent is the best starting point. Both tools are designed to support problem formulation and method selection. The web app is especially useful for users who prefer a structured, step-by-step pathway, whereas the AI agent is more flexible for open-ended, miscellaneous questions.

If the goal is to gain fast hands-on experience with how AI algorithms are applied in PE, the case-study notebooks are the most direct entry point. These examples allow users to quickly see how representative AI methods are used for diversified PE tasks, making them particularly suitable for beginners who want to go beyond theory and observe practical implementation details.

If the target is to develop a systematic and deeper understanding of AI for PE, it is recommended to follow the sequence of the full GitHub repository. Users can begin with the foundational modules covering metaheuristic algorithms, ML, ensemble learning, physics-informed ML, reinforcement learning, and agentic AI, and then progress to the downstream PE case studies. Beyond serving as a code base, the GitHub repository also functions as a structured learning curriculum for mastering AI in the PE context.

By combining code resources in GitHub, interactive decision support from the web app, and AI-assisted guidance by the AI agent, they provide multiple entry points for learners with different backgrounds, goals, and levels of prior experience.

Section IV. Conclusion and Outlook

AI is opening new perspectives for solving PE problems and is becoming increasingly important across PE lifecycle phases. To bridge the gap that current PE education still rarely introduces AI as a systematic engineering toolbox, UA was first to launch a graduate-level course on the fundamentals of AI for PE in Fall 2025, combining AI conceptual instruction, vibe programming, and project-based learning with PE applications. The early outcomes are encouraging: students with little or no prior AI background developed practical AI models and customized generative AI agents (PE-GPT) for PE design automation. Complementing the course, open-source code materials in Jupyter Notebooks, an algorithm selector web app, and a customized AI agent have also been developed to support broader learning and adoption.

Based on these ongoing educational efforts and toward preparing the next generation of AI-ready PE engineers, we would like to make a clarion call to the entire PE community.

To students, now is the right time to learn AI for PE. AI is no longer peripheral to PE research and practice; it is becoming an enabling skill for future scientific discovery and technology innovation. Students can use the provided open-source code materials as an entry point, study the underlying concepts rather than only the final results, and leverage generative AI tools to reflect on, extend, and customize AI for their own PE scenarios. The goal is not merely to use AI tools, but to develop the judgment needed to adapt them responsibly and effectively to PE domain-specific problems.

To educators, AI should be progressively integrated into PE education at multiple levels. Existing PE-oriented courses can include AI-related topics, sections, or modules to expose students to modern data-driven methods without displacing core PE fundamentals. At the same time, new courses dedicated to AI for PE should be developed to provide systematic PE-specific AI training. More broadly, PE curricula should be enriched with foundational AI education so that students can build fluency in both physical reasoning and AI methods. In this way, AI becomes a natural extension of PE training rather than an external add-on.

To industry practitioners, workforce upskilling is equally important. For current engineers, AI-in-PE bootcamps, short courses, and professional training programs can help close the growing skills gap. Companies should also actively recruit PE engineers with AI expertise and initiate AI-driven projects to address PE industry challenges. In parallel, industry can make a major contribution to education and innovation by open-sourcing non-confidential datasets, AI algorithms, and other resources, which would accelerate progress across the PE community.

To decision makers and government agencies, dedicated support is needed to scale this transition. New funding programs should be established for teaching tools, course development, curriculum innovation, and educational infrastructure at the intersection of AI and PE. In addition, agencies can support consortium forming that unites universities, industry, and national labs to co-develop the next generation of AI-integrated PE education. Earlier outreach is also important: K–12 programs that connect AI with electrical engineering can help broaden awareness and build future talent pipelines. Policies, awards, and initiatives that encourage open-source data collection, curation, and publication would further strengthen the ecosystem and make AI education in PE more practical, inclusive, and sustainable.

Looking ahead, the weaving together of AI for PE and PE for AI is more than an emerging topic: it is a defining shift that will shape the future of the field. As AI becomes woven into how PE systems are modelled, designed, controlled, and maintained, the opportunity ahead is both promising and inspiring. By acting now, through education, collaboration, and shared commitment, we can empower a new generation of PE engineers who are fluent in both PE insights and AI intelligent tools, and who will lead the next era of innovation in smart electrification, energy systems, and power conversion.

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